

Chained LQR Funnels for Landing with Certified Deadline-Aware Contingency

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Abstract—FOR REDUCED ORDER MODEL OF AERIALS? We extend steering with landing contingencies to initial states outside the candidate targets’ terminal regions using graphs of overlapping Lyapunov funnels. Free polytopes propose routes, while a convex program certifies each stone’s six-dimensional position–velocity region and admissible feedback; full-state containment then supplies finite-time handoff thresholds and route time bounds. A nested combinatorial filter preserves access to at least r of p target branches while steering along a selected route. Attaching a shrinking allowed Lyapunov level to each certified edge turns the remaining deadline into a differentiable margin encoding funnel membership, time feasibility, and progress. The invariance results are conditional on joint feasibility of simultaneously critical rows. Graph updates are treated as discrete events because pruning can remove reachability at unchanged vehicle state; restart checks and local repair prevent such jumps from being hidden inside the continuous-time proof.

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Index Terms—contingency planning, control Lyapunov functions, control barrier functions, funnel libraries, safe landing.

I. BACKGROUND AND MOTIVATION WILL REWRITE

Consider an aerial vehicle descending toward an unfamiliar environment with several potential landing sites. During descent, an obstacle may be revealed, wind may push the vehicle away from its nominal path, or sensing may degrade. Committing to a single site can leave the vehicle without a safe alternative if that site becomes unreachable. The contingency objective is therefore to preserve access to at least r of p candidate targets while progressing toward a selected site.

Lishkova and Ong’s steering-with-contingency filter [1] provides a framework for maintaining such alternatives. Each target has a certified region of attraction, and the filter modifies a nominal control command to preserve membership in the region where at least r target certificates remain valid. Its application to distant, geographically separated landing sites nevertheless presents two difficulties.

First, the guarantee requires the initial state to lie inside this combinatorial region of attraction. When the targets and their certified regions are separated, the vehicle may begin outside all of them. The filter can preserve existing certificates, but it does not construct the routes needed to extend them to a distant initial state. Second, preserving alternatives does not necessarily bring a preferred target into reach. For example, a 2-of-3 filter may maintain certificates for sites A and B indefinitely while the desired site is C . Steering toward C remains

relaxed outside its certified region, so maintaining contingency alone does not guarantee that its certificate becomes available.

We address these difficulties by constructing a directed graph of locally certified “stepping stones” connecting the initial state to candidate landing sites, as illustrated in Fig. 1. Each stone is a Lyapunov funnel with an admissible local controller and a certified transition to a successor. Consecutive stones overlap geometrically, and their full-state certificates ensure that a transition enters the next funnel in bounded time. This extends target reachability along routes around obstacles, beyond the region covered by a single target-centered certificate.

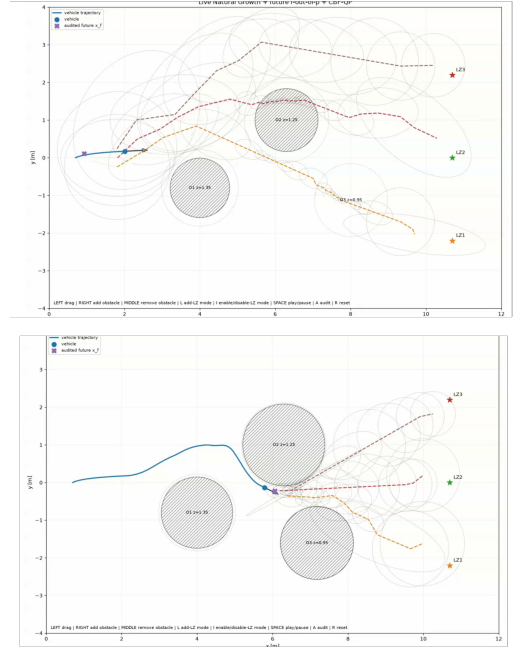


Fig. 1. Landing with contingency through branches of certified stepping stones. The graph connects the initial state to candidate landing sites, while the filter preserves access to multiple target routes during nominal steering.

The graph also gives a combinatorial reach-avoid interpretation. A branch containing p_{branch} usable stones represents a 1-of- p_{branch} region. Membership in any of its stones provides a certified continuation to the terminal target, provided the corresponding transition conditions are enforced. Across the p target branches, membership in at least r branch regions therefore defines an r -of- p combinatorial reach-avoid tunnel.

The state may belong to different overlapping stones from these branches. The construction does not require a single stone to be shared by all r branches.

A contingency filter supervises the nominal controller over this structure. The nominal controller, such as MPC, can use points in successive overlaps as waypoints. The filter restricts its commands to preserve the required target options while enforcing local progress along the selected branch. We first formulate this requirement geometrically, using membership in the certified branch regions. Geometric membership establishes the availability of a route, but does not establish that the target can be reached within a remaining time budget. We therefore also construct a time-based filter using explicit worst-case transition times and a graph-based bound on the remaining travel time to each target.

The present formulation uses a reduced-order linear model and begins with a known, static obstacle map. Its guarantees require successful graph certification, a valid initial contingency condition, and joint feasibility of the steering and filtering constraints. Online map changes must then be treated explicitly. Removing an invalid stone can eliminate a route, make cached travel times stale, and abruptly change the filter's margins even when the vehicle state is unchanged. We examine which assumptions of the continuous filter fail under these updates and provide discrete validity checks, local graph repair, and an operating policy when the requested contingency cannot be maintained.

A. Related Work

The proposed framework builds on the combinatorial filter of Lishkova and Ong [1], including its vanishing-slack mechanism. We apply this mechanism to intermediate route certificates that extend reachability toward distant initial states. The additional graph structure supplies local steering objectives, certified handoffs, and travel-time bounds for the filtering problem.

The construction also draws on LQR-trees [3] and IRIS [4]. LQR-trees motivate the composition of local feedback certificates, while IRIS provides obstacle separation and region inflation. Here, common linear dynamics simplify local certification, and directed growth concentrates on routes from the initial state to specified targets. This avoids constructing a broad cover of the state space, but may miss feasible corridors or produce more conservative routes. Geometric overlap assists construction, while full-state containment remains necessary to certify each handoff.

Two pieces of the construction echo classical graph search, on a graph whose edges are certificates rather than given weights. Algorithm 1's directed, heading-biased expansion toward a target plays the role of A*'s heuristic-guided expansion [5], except that an "edge" here is not looked up but produced and certified: a step can fail for dynamical reasons (an infeasible SDP, an unreachable threshold) that a purely geometric search would never encounter, so expansion and verification are the same step rather than two. The repair procedure of Sec. V-B is the same idea behind D*/D*-Lite [6],

[7]: after an edge changes, reuse untouched cost-to-go values and propagate the update only to the affected predecessors rather than resolving the whole graph. What differs is what an edge cost is allowed to mean. D*/D*-Lite treat edge weights as externally given numbers that can change in either direction between updates; here every $\bar{\tau}_e$ is itself a certified worst-case transfer time tied to one specific SDP-verified funnel, so under pure deletion it can only ever increase at an affected node (Sec. V-A), and a "repaired" edge is not re-weighted but re-certified from scratch by Algorithm 1.

Mixed-integer planning and Hamilton–Jacobi reachability offer alternative approaches to multi-target reach-avoid problems [10], [11]. Although our nominal dynamics are linear, obstacle avoidance and the requirement to retain several destinations remain nonconvex and combinatorial. Our objective is to expose this structure through a compact graph of local certificates and a small online supervisory QP. We do not claim global coverage or globally minimum-time routes.

B. Contributions

This paper makes the following contributions.

- 1) We develop a seeded, directed construction of chained Lyapunov funnels for a reduced-order linear model. The construction combines prescribed geometric overlap with full-state handoff tests and explicit bounds on transition and target arrival times.
- 2) We formulate a contingency filter as a two-level nested combinatorial certificate: one-of-many over the stones of a branch, and r -of- p over the branches. The rows are imposed on the stones, so the pivot argument of the underlying combinatorial framework applies with only the pivot redefined, and the number of constraints stays equal to the number of primitive certificates rather than growing with $\binom{p}{r}$. We also give the offline condition on the graph that makes the resulting rows jointly satisfiable along a primary route.
- 3) We give a deadline-restricted version in which each certified edge carries a shrinking allowed Lyapunov level. A single normalized margin then encodes funnel membership and remaining-time feasibility simultaneously, supplies an explicit progress requirement, and is piecewise C^1 with finitely many scheduled events. This removes the differentiability assumption that the finite-horizon reach-avoid form of the baseline filter requires but does not generally satisfy.
- 4) We identify how graph updates invalidate the assumptions underlying continuous filtering. We provide discrete checks, local routing repair, and explicit responses to contingency loss.

Offline construction and online filtering algorithms support implementation around a nominal waypoint-tracking controller. Validation is planned in simulation, software-in-the-loop, and hardware-in-the-loop on Crazyflie platforms. Future extensions include higher-fidelity nonlinear models, broader graph coverage, robustness to model and mapping uncertainty,

and perception-driven updates using onboard cameras or point clouds.

C. Summary, and Why This Is Needed

Rather than certifying one large region per landing site, we lay a chain of small stepping stones through free space, from the vehicle's start to each candidate site. Each stone carries its own local guarantee: once inside it, a local steering law provably drives the vehicle across it in bounded time and into the next. Doing this per site gives p overlapping “trail systems” through the same airspace, each a certified collision-free route to its own target.

Contingency is then a statement about these trails. The naive version asks how many trails the vehicle currently stands on (Sec. IV-A); the better version (Sec. IV-B) asks *if I committed to this trail now, worst case, would I still have time to finish it?* Each stone's decay rate converts into a crossing time by (4), so these sum offline into a routing table — a per-site “fuel gauge” read live in flight — and contingency becomes: keep at least r of the p gauges non-negative. Because growth is directed rather than exhaustive, corridors are not globally time-optimal; convoluted branches are never grown, and the guarantee is certified sufficiency, not optimality.

Why this indirection is needed at all, given (1) is linear: the obstacle is not the dynamics but the requirement — “at least r of p conditions hold” is combinatorial. Encoding it in the MPC cost or constraints turns a convex program into a mixed-integer one enumerating up to $\binom{p}{r}$ subsets, unacceptable in a receding-horizon loop and fragile besides, since the active subset changes between solves. Exact Hamilton–Jacobi–Bellman (HJB) reachability avoids the combinatorics but reintroduces the curse of dimensionality: a grid over the six-dimensional state plus a time-budget coordinate is impractical to recompute when the map changes, and grid-based HJB solvers are themselves approximations that do not exactly satisfy the viscosity-solution theory justifying them. We instead pay a one-time offline cost, leaving an online QP whose contingency rows scale with the number of primitive stone certificates rather than with the combinatorial number of r -target subsets.

II. PRELIMINARIES WILL REWORK, FOR ME: SAY SOMEWHERE REDUCED ORDER MODEL

A. Model and Notation

The reduced-order translational model is

$$x = \begin{bmatrix} p \\ v \end{bmatrix}, \quad \dot{x} = Ax + Ba, \quad A = \begin{bmatrix} 0 & I_3 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ I_3 \end{bmatrix}, \quad (1)$$

where $p, v \in \mathbb{R}^3$ and $a \in \mathcal{U} := \{a : |a_\ell| \leq \bar{a}_\ell\}$. The scalar p also denotes the number of targets, indexed by j , and $r \leq p$ the required number of available targets. Inflating obstacles by the vehicle radius and a geometric margin gives $\mathcal{O}_q$; eroding the workspace gives W^+ . Their free-position domain is $\Omega_f = W^+ \setminus \bigcup_q \mathcal{O}_q$. We write $\|z\|_M^2 = z^\top M z$ and $\dot{\phi} = L_f \phi + L_g \phi a$, with $f(x) = Ax$, $g(x) = B$.

B. Control Lyapunov Functions and Time Bounds

A positive-definite C^1 function V about an equilibrium x^* is a local control Lyapunov function (CLF) if, throughout a neighborhood, an admissible input satisfies $\dot{V} \leq -\alpha(V)$ for a class- $\mathcal{K}$ function α . Here $V = \|x - x^*\|_P^2$, $P \succ 0$, and we enforce

$$L_f V + L_g V a \leq -kV, \quad k > 0. \quad (2)$$

Lemma 1 (Exponential comparison). *If (2) holds almost everywhere along an absolutely continuous trajectory, then*

$$V(x(t)) \leq V(x(t_0))e^{-k(t-t_0)}. \quad (3)$$

Proof. $\frac{d}{dt}(e^{k(t-t_0)}V) = e^{k(t-t_0)}(\dot{V} + kV) \leq 0$. $\square$

Consequently, a sublevel set $\mathcal{R}(\rho) = \{x : V(x) \leq \rho\}$ is forward invariant and certifies attraction if (2) remains feasible throughout it. Moreover, by solving $V_0 e^{-k\tau} = \varepsilon$, the time to drive V down to any threshold ε , can be bounded as follows for $V(x(t_0)) > \varepsilon$:

$$\tau \leq \frac{1}{k} \log \frac{V(x(t_0))}{\varepsilon} \quad (4)$$

This positive threshold gives finite arrival time, whereas convergence to the equilibrium itself is asymptotic. We write

$$\tau_{\text{dec}}(V; k, \varepsilon) := k^{-1} \log \max\{1, V/\varepsilon\} \quad (5)$$

for the same bound, with value zero when $V \leq \varepsilon$.

Remark 1. will rewrite Under a controller enforcing $\dot{V} \leq -kV$, define the time-dependent funnel cross-section

$$\mathcal{F}(t) := \{x : V(x) \leq ce^{-k(t-t_i)}\}, \quad t \geq t_i. \quad (6)$$

Trajectories starting in $\mathcal{R}(c)$ at time t_i remain inside $\mathcal{F}(t)$. Stacking these progressively smaller cross-sections along the time axis forms a narrowing tube toward the local target, giving the funnel picture. We use *level set* when discussing the region's geometry and *funnel* when discussing its associated controller and certified convergence. When referring to its fixed outer region, we write $\mathcal{F} := \mathcal{F}(t_i) = \mathcal{R}(c)$.

C. Control Barrier Functions

For $\mathcal{C} = \{x : h(x) \geq 0\}$, a C^1 function h is a control barrier function (CBF) when an admissible input satisfies

$$L_f h + L_g h a \geq -\alpha(h) \quad (7)$$

on a neighborhood of $\mathcal{C}$, with α extended class- $\mathcal{K}$. Under the usual feedback and solution regularity conditions, (7) makes $\mathcal{C}$ forward invariant [8]. For $\alpha(h) = \gamma h$, comparison gives $h(t) \geq h(t_0)e^{-\gamma(t-t_0)}$. CLF and CBF conditions are affine in a , so they can constrain a quadratic-program (QP) filter around a nominal controller. Their joint feasibility must still be established.

D. Steering with Combinatorial Contingency

For each target j , let V_j be a CLF with certified region of attraction $\mathcal{R}_j = \{x : V_j(x) \leq c_j\}$. Following the notation of [1], define the reachability margin and combinatorial pivot by

$$\begin{aligned} h_j^{\text{ROA}}(x) &:= c_j - V_j(x), \\ \tilde{h}_r(x) &:= \max^{(r)}\{h_1^{\text{ROA}}(x), \dots, h_p^{\text{ROA}}(x)\}, \end{aligned} \quad (8)$$

where $\max^{(r)}$ returns the r th largest value. Thus, $\tilde{h}_r(x) \geq 0$ holds exactly when at least r of the p certified regions contain the state. The *pivot* is the weakest margin among the best r options. Its realizing targets form $\mathcal{P}_r(x) = \{j : h_j^{\text{ROA}}(x) = \tilde{h}_r(x)\}$, including all ties, and may change as the state moves.

With $j^\dagger$ the selected primary target, the filter solves

$$\min_{a, \omega \geq 0} \quad \frac{1}{2} \|a - a_{\text{nom}}\|_2^2 + p_\omega \omega^2 \quad (9)$$

subject to the input bounds $-\bar{a} \leq a \leq \bar{a}$ and the primary CLF-based steering condition

$$\begin{aligned} L_f V_{j^\dagger} + L_g V_{j^\dagger} a &\leq -k_V V_{j^\dagger} \\ &\quad + \omega \text{ReLU}(-h_{j^\dagger}^{\text{ROA}}). \end{aligned} \quad (10)$$

and a contingency reach-avoid constraint for $j = 1, \dots, p$,

$$\begin{aligned} L_f h_j^{\text{ROA}} + L_g h_j^{\text{ROA}} a &\geq -k_V h_j^{\text{ROA}} \\ &\quad - \omega \rho(h_j^{\text{ROA}} - \tilde{h}_r), \end{aligned} \quad (11)$$

where $\rho(s) = k_\rho s^2$, $\text{ReLU}(s) = \max\{0, s\}$, and $k_V, k_\rho, p_\omega > 0$. The shared variable ω permits relaxation without weakening the current pivot condition. Inside the primary region, $\text{ReLU}(-h_{j^\dagger}^{\text{ROA}}) = 0$, so (10) enforces unrelaxed CLF decay. Likewise, $\rho(0) = 0$ removes all slack from (11) for every target currently realizing the pivot. Protection therefore follows the live order statistic, without fixing the protected targets in advance. Other rows retain relaxation, although joint feasibility must still hold. Under the framework's regularity and feasibility conditions, initialization with $\tilde{h}_r \geq 0$ ensures that at least r target certificates remain valid.

Remark 2 (Steering does not by itself sustain contingency). Suppose stones $\mathcal{A}, \mathcal{B}, \mathcal{C}$ must be traversed in order to reach a target in $\mathcal{C}$, and the requested level is $r = 2$. At a state in $\mathcal{A} \cap \mathcal{B}$ the contingency condition is satisfied, and nothing in the contingency rows pushes the vehicle toward $\mathcal{B} \cap \mathcal{C}$. Progress is supplied only by the primary steering row, and preserving alternatives can delay or prevent arrival. Two consequences are used later: the construction must guarantee that the *primary* route lies in the certified r -of- p region, so that the steering row and the contingency rows are jointly satisfiable (Sec. III-E); and deadline feasibility of retained alternatives must be encoded explicitly, which the deadline-restricted filter of Sec. IV-B does through shrinking certified levels.

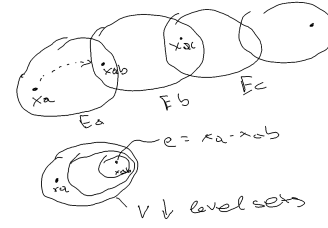


Fig. 2. Stepping-stone traversal. Displayed regions are position projections of full-state funnels; switching is certified in the full state.

III. PROPOSED GRAPH GROWTH IN A KNOWN STATIC MAP

A. Certified stepping stones

Definition 1 (Search region and certified region). A *free polytope* is a bounded set $\mathcal{P}_i = \{p \in \mathbb{R}^3 : n_{i\ell}^\top p \leq b_{i\ell}, \ell = 1, \dots, m_i\} \subset \Omega_f$ of obstacle-free positions. A *stone*, or *funnel*, is a full-state set $\mathcal{F}_i \subset \mathbb{R}^6$ equipped with a triple $(\hat{x}_i, K_i, k_i)$ of aim, feedback gain, and certified decay rate, satisfying (22) below.

A certified stone is instead a full-state funnel $\mathcal{F}_i \subset \mathbb{R}^6$ with aim $\hat{x}_i$, feedback K_i , and rate k_i satisfying (22). The vehicle enters a stone, enforces its CLF decay, and switches when a positive level $V_i \leq \varepsilon_{ik}$ is contained in the successor funnel. Hence no stop at an aim is required. Chaining these local guarantees yields a collision-free route with an explicit finite-time bound.

B. Stage One: Proposing a Direction and a Polytope

1) *Choosing a growth direction*: For growth origin $z \in \text{int } \mathcal{P}_i$ and target/waypoint g , start from $d_0 = (g - z)/\|g - z\|$. A unit heading d is admissible when

$$\text{LoS}(z, d) = \mathbf{1}[z + sd \in \Omega_f \quad \forall s \in [0, \ell_{1a}]]. \quad (12)$$

If d_0 is blocked, sweep a fan

$$d(\theta, \phi) = \cos \theta d_0 + \sin \theta (\cos \phi u + \sin \phi w), \quad (13)$$

for an orthonormal (d_0, u, w) and bounded θ , followed by randomized admissible headings if needed. Failure within the search budget backtracks the branch; completeness is not claimed.

2) *Seeded overlap*: Let r_d be the first ray–face intersection distance,

$$r_d = \min_{\ell: n_{i\ell}^\top d > 0} \frac{b_{i\ell} - n_{i\ell}^\top z}{n_{i\ell}^\top d}, \quad c_k = z + \beta r_d d, \quad 0 < \beta < 1. \quad (14)$$

Then $c_k \in \text{int } \mathcal{P}_i$; any full-dimensional child polytope with c_k in its interior therefore has interior geometric overlap with the parent. This is only a candidate connection; Stage Two certifies its dynamical traversability.

3) *Constructing the child polytope*: IRIS [4] normally alternates two convex steps: (a) given a current region, generate a separating hyperplane against every obstacle that intersects it; and (b) given the resulting polytope, solve a maximum-volume-inscribed-ellipsoid SDP, then feed that ellipsoid back into step (a) to grow the polytope further, repeating a handful of times until the volume stops improving. We use only step (a), once, per seed – never the ellipsoid-inflation SDP of step (b), and never the alternation. The polytope produced this way is scaffolding: it is discarded the moment Stage Two below has certified a full-state region inside it, so there is nothing to gain from spending an SDP to inflate a good *position-only* ellipsoid, when the very next step replaces it with a six-dimensional position-velocity ellipsoid anyway (Remark 10 makes this comparison precise once Stage Three has introduced the last piece of the comparison).

Concretely, for each relevant convex inflated obstacle $\overline{\mathcal{O}}_q$ near the seed c_k , and a fixed metric $M \succ 0$, compute

$$y_q = \arg \min_{p \in \overline{\mathcal{O}}_q} (p - c_k)^\top M (p - c_k), \quad n_q = \frac{M(y_q - c_k)}{\|M(y_q - c_k)\|}.$$

Here y_q is the point of $\overline{\mathcal{O}}_q$ closest to the seed in the metric M – a small convex program in general, closed-form when $\overline{\mathcal{O}}_q$ is a sphere or a simple polytope, and cheap either way since it involves only one obstacle and one seed point, not an optimization over the whole free space. Because $\overline{\mathcal{O}}_q$ is convex, it lies entirely on one side of the tangent plane at y_q , so $\overline{\mathcal{O}}_q \subset \{p : n_q^\top p \geq n_q^\top y_q\}$ and the complementary half-space

$$n_q^\top p \leq n_q^\top y_q - \delta_q, \quad 0 < \delta_q < n_q^\top (y_q - c_k), \quad (15)$$

is certified obstacle-free, with δ_q a strict margin chosen small enough to keep the seed c_k itself feasible. This is the precise sense in which the construction *proves* $\mathcal{P}_k \subset \Omega_f$ rather than assuming it: each retained half-space in (15) excludes exactly one obstacle by a one-line convexity argument, workspace bounds are added the same way, and $\mathcal{P}_k$ is their intersection. Every obstacle that could intersect the candidate region before it shrinks to its final size must be included in this pass; distance-based pruning of “irrelevant” obstacles is only admissible if it provably cannot exclude an intersection. Computationally this whole step is one closest-point query per nearby obstacle – for the handful of obstacles relevant to one seed this is sub-millisecond to a few milliseconds on commodity hardware, far cheaper than the shape SDP of Stage Two that follows it and orders of magnitude cheaper than iterating full IRIS’s ellipsoid step.

C. Stage Two: Certifying the Dynamics

Inside the proposed polytope we now choose the certified region, its feedback, and the level at which it hands off. Three quantities must be selected: where convergence is directed, how fast decay is demanded, and what shape the certified region takes.

1) *Where convergence is directed*: The feedback drives the state toward an aim. Placing that aim in the overlap with the intended successor makes local convergence and route progress the same event:

$$\hat{x}_i = [\hat{p}_i; 0], \quad \hat{p}_i \in \text{int}(\mathcal{P}_i \cap \mathcal{P}_k), \quad (16)$$

for which the overlap’s Chebyshev center is a candidate. At the final edge the aim must additionally lie strictly inside the full-state terminal funnel with its required margin; overlap of position projections does not establish this. All aims on a candidate path are fixed before any shape is optimized.

Remark 3 (Why the aim velocity is zero). Under $a = -K_i(x - \hat{x}_i)$ the error $e_i = x - \hat{x}_i$ obeys

$$\dot{e}_i = (A - BK_i)e_i + [\hat{v}_i; 0]. \quad (17)$$

Only $\hat{v}_i = 0$ makes the aim an equilibrium of the double integrator. A nonzero $\hat{v}_i$ leaves a constant forcing term, which contributes a linear term to $\dot{V}_i$ and defeats quadratic decay near the aim; acceleration feedforward does not remove the resulting position offset. This is a restriction on the *reference*, not on the vehicle, which hands off at the strictly positive level ε_{ik} of (30) and is never required to come to rest. A cruise-velocity reference would require a moving aim and a different transfer certificate.

2) *How fast decay is demanded*: Fix a stabilizing gain K_i , for instance by LQR, and write $A_{cl,i} = A - BK_i$. The demanded decay rate is bounded by the closed loop’s own spectrum,

$$0 < k_i < k_i^{\max} := -2 \max \text{Re } \lambda(A_{cl,i}). \quad (18)$$

The gain, the rate, the aim, the polytope, and the heading are held fixed during the shape solve below. Candidate rates are compared by re-solving and evaluating the resulting transfer bounds: a larger k_i demands faster decay but shrinks the certified region, and it need neither improve transfer time nor retain a feasible entry state. Remark 5 removes the need to fix K_i in advance.

3) *What shape the certified region takes*: Parameterize the certificate by its size matrix rather than by its metric. For symmetric

$$H_i = \begin{bmatrix} H_{i,pp} & H_{i,pv} \\ H_{i,vp} & H_{i,vv} \end{bmatrix} \succ 0,$$

define

$$P_i = H_i^{-1}, \quad \rho_i = 1, \quad V_i(x) = (x - \hat{x}_i)^\top P_i (x - \hat{x}_i), \quad \mathcal{F}_i = \{x : V_i(x) \leq 1\}. \quad (19)$$

Normalizing the outer level to one loses no ellipsoid, since $H_i = \rho_i P_i^{-1}$ in level-set notation and rescaling the level is the same operation as rescaling H_i : metric and retained size are now chosen together. A supplied terminal certificate is normalized the same way. The normalization also makes $\rho_i - V_i$ and $1 - V_i/\rho_i$ coincide, so the margins compared by the pivots of Sec. IV are dimensionless and consistently scaled across cells. Consequently $\rho_i \equiv 1$ for every certified funnel

used in this paper, with no exceptions; every later appearance of ρ_i (or $\rho_{j,i}$) is kept purely to mark which quantities are conceptually tied to the outer level, not because it ever takes a value other than one. A reader who prefers may simply read $\rho_{j,i} = 1$ everywhere below.

(for me: just go this is the direction, this is the point chosen,, this rate, here is the socp)

With $\Delta_{i\ell} = b_{i\ell} - n_{i\ell}^\top \hat{p}_i > 0$ the signed distance from the aim to face ℓ , and fixed buffers $0 < \gamma_g, \gamma_u < 1$, the shape is chosen by

$$\begin{aligned} \max_{H_i \succ 0} \quad & \log \det H_i \\ \text{s.t.} \quad & A_{\text{cl},i} H_i + H_i A_{\text{cl},i}^\top + k_i H_i \preceq 0, \\ & n_{i\ell}^\top H_i n_{i\ell} \leq \gamma_g \Delta_{i\ell}^2, \quad \ell = 1, \dots, m_i, \\ & K_{i,q} H_i K_{i,q}^\top \leq \gamma_u \bar{a}_q^2, \quad q = 1, 2, 3, \end{aligned} \quad (20)$$

where $K_{i,q}$ is row q of K_i . The three constraint families correspond one to one with the three ways a candidate region can fail, and each admits a direct reading.

Decay. Congruence of the first inequality with P_i gives $A_{\text{cl},i}^\top P_i + P_i A_{\text{cl},i} + k_i P_i \preceq 0$, which is precisely $\dot{V}_i \leq -k_i V_i$ under $a = -K_i(x - \hat{x}_i)$, everywhere in $\mathcal{F}_i$.

Clearance. Marginalizing V_i over the velocity block at fixed position – a Schur complement of $P_i = H_i^{-1}$ with respect to its vv block – shows the position projection of (19) is exactly the ellipsoid

$$\Pi_p \mathcal{F}_i = \{p \in \mathbb{R}^3 : (p - \hat{p}_i)^\top H_{i,pp}^{-1} (p - \hat{p}_i) \leq 1\}, \quad (21)$$

with support $\sqrt{n^\top H_{i,pp} n}$ in direction n . This is used twice, and only twice, in the whole construction: it is why the clearance row below controls the funnel's true footprint rather than a conservative proxy, and it is what Algorithm 1's terminal check $\mathcal{P}_{i_n} \cap \Pi_p \mathcal{F}_{j,T} \neq \emptyset$ actually tests. It is exact but one-directional: a position landing inside (21) does not certify the full state lies in $\mathcal{F}_i$ at an arbitrary velocity, so Fig. 2 displays this projection, not the certified set. The second constraint family states that the projection does not reach face ℓ , with the factor γ_g retaining a buffer.

Actuation. The largest magnitude of feedback channel q over the funnel is $\sqrt{K_{i,q} H_i K_{i,q}^\top}$, so the third family states that the certifying feedback never leaves the input box, again with a buffer.

All three are affine in H_i and the objective is concave, so (20) is a convex program solved once, with no inflate-and-test iteration. The cross block $H_{i,pv}$ is left free. A nonzero cross block tilts the certified region so that high speed along the corridor is admitted while motion toward a face is restricted. No region defined in position space alone can express that distinction. Together the constraints give

$$\Pi_p \mathcal{F}_i \subset \mathcal{P}_i \subset \Omega_f, \quad (22)$$

$$\mathcal{U}_i(x) := \{a \in \mathcal{U} : L_f V_i + L_g V_i a \leq -k_i V_i\} \neq \emptyset, \quad x \in \mathcal{F}_i,$$

the second condition witnessed by $-K_i(x - \hat{x}_i)$. We write the admissible-decay set as $\mathcal{U}_i(x)$, deliberately distinct from the feedback gain K_i ; the two are unrelated objects (a set of inputs

versus a matrix) that would otherwise be typographically hard to tell apart. The buffers provide geometric and actuation margin, not robustness to unmodeled disturbances or actuation error; certifying that margin under a bounded disturbance model is left to future work, and every result below assumes the exact model of Assumption 1.

Remark 4 (Optional structural constraints). Two further requirements, when present, are additional affine rows in the same program rather than separate stages, and may be omitted otherwise. To retain elongation along the growth heading d , fix $R_d \in SO(3)$ with first column d and impose

$$\begin{aligned} H_{i,pp} &= R_d \text{diag}(\sigma_1, \sigma_2, \sigma_3) R_d^\top, \\ \sigma_1 &\geq \sigma_2, \quad \sigma_1 \geq \sigma_3, \end{aligned} \quad (23)$$

which aligns a longest projected semi-axis with d and preserves convexity. To respect explicit velocity limits $|v_q| \leq \bar{v}_q$, impose

$$(H_{i,vv})_{qq} \leq \bar{v}_q^2, \quad q = 1, 2, 3, \quad (24)$$

which is exact because $\max_{x \in \mathcal{F}_i} |v_q| = \sqrt{(H_{i,vv})_{qq}}$. Admissible velocities remain restricted by clearance, actuation, and the decay condition even when (24) is omitted. Imposing (23) on the certificate rather than on a position-space proxy is what preserves the anisotropy: a directed region inflated in position space and then lifted contributes only its worst-aligned direction to the retained level.

Remark 5 (Gain co-design). Fixing K_i in advance is unnecessary. Substituting $Y_i = K_i H_i$ turns the first row of (20) into $A H_i - B Y_i + H_i A^\top - Y_i^\top B^\top + k_i H_i \preceq 0$ and the actuation rows into

$$\begin{bmatrix} \gamma_u \bar{a}_q^2 & Y_{i,q} \\ Y_{i,q}^\top & H_i \end{bmatrix} \succeq 0, \quad q = 1, 2, 3,$$

both jointly convex in (H_i, Y_i) , with the certifying feedback recovered as $K_i = Y_i H_i^{-1}$. Since the LQR gain is feasible for this program, co-design can only enlarge the retained region. The rate k_i must still be fixed or searched over, because $k_i H_i$ enters bilinearly otherwise.

Remark 6 (Well-posedness). The feasible set of (20) is bounded, so a maximizer exists: a bounded $\mathcal{P}_i$ has face normals spanning $\mathbb{R}^3$, which bounds $H_{i,pp}$, and if $K_i = [K_{i,p} \ K_{i,v}]$ has $K_{i,v}$ nonsingular then the rows of K_i together with the position coordinates span $\mathbb{R}^6$, so boundedness of $v^\top H_i v$ on a spanning set bounds $H_i \succ 0$. Feasibility is not automatic, since the entry constraint below requires H_i to be sufficiently large; its infeasibility is the acceptance test of Remark 9.

4) *Containing the arriving state:* A certified region is of no use unless it contains the state that arrives in it, so containment is imposed as a constraint rather than verified afterwards. For a required state x_{in} and fixed $0 < \eta_i \leq 1$,

$$\begin{bmatrix} \eta_i H_i & x_{\text{in}} - \hat{x}_i \\ (x_{\text{in}} - \hat{x}_i)^\top & 1 \end{bmatrix} \succeq 0, \quad (25)$$

whose Schur complement is exactly $V_i(x_{\text{in}}) \leq \eta_i$ and which is affine in H_i . On a candidate path, every non-root funnel

takes the predecessor's aim as a required entry state, with $\eta_i < 1 - \zeta$ for the margin ζ of (30), which guarantees that a strictly positive handoff threshold can subsequently be found. The threshold test still certifies the whole switching region, including its velocities, rather than the aim alone. Because (25) is a constraint, an infeasible candidate is rejected by a single solve, and the dual identifies the binding face or actuation row.

At the root, $c_0 = p_0 = p(t_0)$ and $\hat{x}_0 = [p_0; 0]$, so the requirement is a condition on the initial velocity,

$$V_0(x(t_0)) = v_0^\top P_{0,vv} v_0 \leq \eta_0 \leq 1, \quad (26)$$

imposed through (25) with $x_{\text{in}} = x(t_0)$, since (26) is not itself affine in H_0 . The first successor must contain $\hat{x}_0$ with the required margin; otherwise the seed, polytope, or certificate must be revised. Initial containment does not by itself certify departure. If several predecessors share one certificate, all of their aims must be included or certified separately.

Collecting the constraints, the shape program for a non-root node, with the optional rows of Remarks 4 and 5 included, takes the form

$$\begin{aligned} & \max_{H_i \succ 0, Y_i, \sigma} \log \det H_i \\ \text{s.t. } & AH_i - BY_i + H_i A^\top - Y_i^\top B^\top + k_i H_i \preceq 0, \quad (\text{decay}) \\ & H_{i,pp} = R_d \text{diag}(\sigma) R_d^\top, \quad \sigma \succ 0, \quad (\text{shape}) \\ & \sigma_1 \geq \sigma_2, \quad \sigma_1 \geq \sigma_3, \\ & \sum_{m=1}^3 \sigma_m (n_{i\ell}^\top R_d e_m)^2 \leq \gamma_g \Delta_{i\ell}^2, \quad (\text{walls}) \\ & \ell = 1, \dots, m_i, \\ & \begin{bmatrix} \gamma_u \bar{a}_q^2 & Y_{i,q} \\ Y_{i,q}^\top & H_i \end{bmatrix} \succeq 0, \quad q = 1, 2, 3, \quad (\text{input}) \\ & (H_{i,vv})_{qq} \leq \bar{v}_q^2, \quad q = 1, 2, 3, \quad (\text{speed}) \\ & \begin{bmatrix} \eta_i H_i & \delta_{\text{in}} \\ \delta_{\text{in}}^\top & 1 \end{bmatrix} \succeq 0, \quad (\text{entry}) \end{aligned} \quad (27)$$

with decision variables $H_i \succ 0$, $Y_i \in \mathbb{R}^{3 \times 6}$ and $\sigma \in \mathbb{R}^3$, and with

$$\begin{aligned} \Delta_{i\ell} &= b_{i\ell} - n_{i\ell}^\top \hat{p}_i > 0, \quad Y_{i,q} = e_q^\top Y_i, \\ \delta_{\text{in}} &= x_{\text{in}} - \hat{x}_i, \quad e_m \in \mathbb{R}^3 \text{ the } m\text{th basis vector.} \end{aligned} \quad (28)$$

The certificate is recovered from the solution as

$$\begin{aligned} K_i &= Y_i H_i^{-1}, \quad P_i = H_i^{-1}, \\ \rho_i &= 1, \quad A_{c1,i} = A - B K_i. \end{aligned} \quad (29)$$

The substitution $Y_i = K_i H_i$ of Remark 5 makes the decay and actuation rows jointly affine in (H_i, Y_i) , so (27) remains convex; because $k_i H_i$ would otherwise enter bilinearly, the rate is held fixed and searched over outside the solve, and infeasibility is the indication that it was set too aggressively. Decay, walls, input, and speed are all, in effect, upper bounds on H_i : any sufficiently small H_i satisfies them trivially, so on their own they would be solved by shrinking to a point.

Entry is the only lower bound, requiring H_i large enough to contain x_{in} at level η_i ; it is therefore the row that makes the program capable of genuine infeasibility rather than a degenerate feasible point, which is the acceptance test of Remark 9. Certification proceeds backward along a candidate path because (31) below requires $\mathcal{F}_k$ to already exist when testing ε_{ik} ; nothing in the shape program itself needs H_k .

5) *Choosing the handoff threshold*: From here on, whenever i and k appear together they denote a stone and its one designated successor, i.e. a directed edge (i, k) of the graph being grown along a single candidate path – never two independently chosen stones. With both shapes fixed, the connection is completed by a level of $\mathcal{F}_i$ that lies entirely inside $\mathcal{F}_k$:

$$\Sigma_{ik} = \{x : V_i(x) \leq \varepsilon_{ik}\} \subset \text{int}(\mathcal{F}_i \cap \mathcal{F}_k), \quad 0 < \varepsilon_{ik} < 1. \quad (30)$$

A necessary preliminary is $V_k(\hat{x}_i) < 1 - \zeta$, which (25) enforces for newly optimized successors and which must be checked separately for a supplied terminal funnel. With $\delta_x = \hat{x}_k - \hat{x}_i$ and a fixed candidate $\varepsilon > 0$, containment is a single quadratic implication and therefore admits a lossless S-procedure test,

$$\exists \lambda \geq 0 : \begin{bmatrix} \lambda P_i - P_k & P_k \delta_x \\ \delta_x^\top P_k & 1 - \zeta - \delta_x^\top P_k \delta_x - \lambda \varepsilon \end{bmatrix} \succeq 0, \quad (31)$$

equivalent to $V_i \leq \varepsilon \Rightarrow V_k \leq 1 - \zeta$. The threshold is obtained by bisecting ε in $(0, 1)$ and retaining a feasible value strictly below one. Condition (31) is affine in P_i rather than in H_i , which is why it forms a separate step after (20) rather than an additional row within it, and why certification proceeds *backward* from the terminal funnel: testing ε_{ik} requires $\mathcal{F}_k$ to exist already. Distinct exit aims require distinct funnel certificates.

Remark 7 (Volume is not the right objective for time). With $\rho_i = 1$ the cached edge bound of (36) is $\bar{\tau}_{ik} = k_i^{-1} \log(1/\varepsilon_{ik})$, so it depends on the shape only through ε_{ik} . A flatter P_i enlarges $\{V_i \leq \varepsilon\}$ at fixed ε , tightening (31) and forcing ε_{ik} down. Maximizing $\log \det H_i$ therefore trades certified travel time for retained volume. No extra machinery is needed to manage this trade-off: solve (27) over a small grid of candidate k_i and keep whichever result gives the best realized $\bar{\tau}_{ik}$ after the bisection on (31).

Remark 8 (Template scaling is a restriction). Fixing a template certificate H^{tpl} satisfying the decay row and optimizing only the scalar in $H_i = \alpha H^{\text{tpl}}$ is a *restriction*: the decay row is homogeneous in H_i and the clearance and actuation rows scale, so every αH^{tpl} is feasible for (20) and $\log \det H_i^* \geq \log \det(\alpha^* H^{\text{tpl}})$ at identical $(K_i, k_i, \hat{x}_i, \mathcal{P}_i)$; adding (23) likewise recovers directed template growth. We use this cheaper restriction only where solving (27) from scratch is too slow to matter, namely online repair in Sec. V-B, and as the offline baseline in Sec. ??.

Remark 9 (Geometric proposal, dynamical acceptance). Stage Two certifies local geometry and dynamics jointly, but a proposed polytope path can still fail its shape, entry, or handoff tests, typically because the available clearance is insufficient

for the required entry velocity under the input bounds. The response is to revise the geometry, aim, gain, or rate, or to reject the candidate. An interior predecessor aim guarantees the existence of some positive handoff threshold, not a large threshold or a short transfer. An accepted connection requires all three of full-state entry, admissible decay, and successor containment. Joint feasibility of the contingency filter is a separate requirement, treated in Sec. III-E.

D. Stage Three: From Decay Rates to Time Bounds

The certified decay rate converts directly into a bound on the duration of a transition, and these bounds compose along a branch, using τ_{dec} from (5).

Proposition 1 (Safe finite transfer). *Let $x(t_i) \in \mathcal{F}_i$ and apply $a(t) \in \mathcal{U}_i(x(t))$ until Σ_{ik} is reached. Then*

$$\begin{aligned} V_i(x(t)) &\leq V_i(x(t_i))e^{-k_i(t-t_i)} \leq 1, \\ t_{i \rightarrow k} - t_i &\leq \tau_{ik}(x(t_i)) := \tau_{\text{dec}}(V_i(x(t_i)); k_i, \varepsilon_{ik}). \end{aligned} \quad (32)$$

The vehicle remains collision-free throughout and enters $\mathcal{F}_k$ at handoff.

Proof. Apply the exponential comparison and (4); safety follows from (22) and successor containment from (30). $\square$

For a finite branch $(i_0, \dots, i_N)$, the generic edge label (i, k) is relabeled by position along the path: (i_m, i_{m+1}) denotes the m -th transition.

Proposition 2 (Chained containment). *For a finite branch $(i_0, \dots, i_N)$ satisfying (22) and the handoff property (30) at each edge, assume initial containment and apply each cell's decay condition until its handoff. Then*

$$x(t) \in \mathcal{F}_{i_m}, \quad t_m \leq t \leq t_{m+1}, \quad m < N, \quad (33)$$

$$p(t) \in \bigcup_{m=0}^N \Pi_p \mathcal{F}_{i_m} \subset \Omega_f, \quad (34)$$

and

$$t_N - t_0 \leq \sum_{m=0}^{N-1} \frac{1}{k_{i_m}} \log \max \left\{ 1, \frac{V_{i_m}(x(t_m))}{\varepsilon_{i_m i_{m+1}}} \right\}. \quad (35)$$

Proof. Each handoff lies in the next funnel, so induction applies Proposition 1 along the branch and sums its bounds. Already-satisfied thresholds permit zero-duration transitions. $\square$

For route selection we deliberately price every edge by its worst case over the whole source funnel. For $e = (i, k)$,

$$\bar{\tau}_e \equiv \bar{\tau}_{ik} = \sup_{x \in \mathcal{F}_i} \tau_{ik}(x) = \frac{1}{k_i} \log \frac{\rho_i}{\varepsilon_{ik}} = \frac{1}{k_i} \log \frac{1}{\varepsilon_{ik}}, \quad (36)$$

where the final equality uses the normalization $\rho_i = 1$. This whole-funnel convention is conservative, but supports arbitrary online attachment and is used by Lemma 3. For the terminal funnel i_T of branch j , choose $\varepsilon_{j,T} > 0$ with $\{V_{j,T} \leq \varepsilon_{j,T}\} \subset \mathcal{T}_j$ and set

$$\bar{\tau}_j^{\text{land}} = \frac{1}{k_{j,T}} \log \frac{1}{\varepsilon_{j,T}}, \quad J_j(i_T) = \bar{\tau}_j^{\text{land}}. \quad (37)$$

Accumulating these costs backward gives the route table

$$J_j(i) = \min_{k \in \text{succ}_j(i)} \{\bar{\tau}_{ik} + J_j(k)\}, \quad (38)$$

with $J_j(i) = +\infty$ for nodes disconnected from the terminal set. It can be computed by a reverse sweep on an acyclic graph or reverse Dijkstra when cross-edges are present.

Online, the measured state gives a tighter bound for the current edge, while the cached suffix bound is retained. Define the available attachments

$$\mathcal{N}_j(x) = \{i : x \in \mathcal{F}_{j,i}, J_j(i) < \infty\}. \quad (39)$$

For the fixed certified graph, local decay feasibility inside every retained funnel is already guaranteed by (22) and need not be repeated in (39). The certified target-level time-to-go is

$$T_j^{\text{cert}}(x) = \min_{\substack{i \in \mathcal{N}_j(x) \\ k \in \text{succ}_j(i)}} \{\tau_{ik}(x) + J_j(k)\}, \quad (40)$$

where

$$\tau_{ik}(x) = \tau_{\text{dec}}(V_{j,i}(x); k_{j,i}, \varepsilon_{ik})$$

is the live certified time remaining on the current edge. A zero-cost sink is added after the terminal funnel; $T_j^{\text{cert}} = 0$ in $\mathcal{T}_j$ and $+\infty$ when unattached. Thus T_j^{cert} summarizes the best currently certified route to target j and is used for commitment and discrete validity checks; the continuous deadline filter below is constructed at the primitive edge level.

Remark 10 (Relation to IRIS and LQR-Trees). The three stages above take exactly one component from each of the two constructions cited in Sec. I-A and change what each one certifies. From IRIS we take only step (a), the one-shot separating-plane polytope of Sec. III-B; we deliberately drop step (b), its own ellipsoid-inflation SDP, because the polytope is scaffolding, discarded once built, and skipping step (b) costs nothing: the position-space footprint of the eventual six-dimensional certificate is itself an exact ellipsoid, (21), centered at the aim rather than at the seed that step (b) would inflate around, so nothing is gained by first maximizing a position-only ellipsoid that Stage Two's program immediately supersedes and re-centers anyway. In its place we solve, once, the six-dimensional position-velocity program (27) – the role IRIS's ellipsoid step and LQR-Trees' sums-of-squares verification both play, made convex instead of bilinear because the plant (1) is linear rather than the general nonlinear systems LQR-Trees targets. Certifying in the full state rather than in position alone is not a cosmetic change: it is what buys the free cross block $H_{i,pv}$, and hence the corridor-aligned tilt (fast along the direction of travel, tight toward a wall) that no position-only region – from IRIS, from an inscribed ellipsoid, or from any other purely geometric construction – can express, because velocity is simply not a coordinate such a region has. Funnel-to-funnel containment, which LQR-Trees checks with a bilinear SOS program, is checked here with the closed-form S-procedure LMI (31), again because linearity turns a generally non-convex certification into a convex one. What neither IRIS nor LQR-Trees provides at all is the Bellman

time-to-go table J_j of (38): it exists only because Sec. IV-B needs a certified, composable arrival-time bound to build a deadline filter, and it is this table, not the funnels themselves, that turns a library of verified regions into a route with a time budget.

E. Obtaining the Contingency Condition

Each target branch is certified independently. Along any intended primary route we additionally require its handoff regions to lie in the geometric r -of- p set and require the simultaneously critical filter rows to admit a common input. These are distinct tests: overlap supplies common state membership, while Proposition 3 below supplies control compatibility.

1) *Cross-branch contingency check*: Run Algorithm 1 once per target, then test per edge of the intended primary route that $\Sigma_{ik} \subset \mathcal{C}_r^{\text{geo}}$, with $\mathcal{C}_r^{\text{geo}}$ from (41), and re-seed or reject an edge that fails. This one check is deliberately the only mechanism used, whether or not two branches happen to share the same underlying polytope $\mathcal{P}_i$: sharing a position-space polytope does not by itself make two independently certified six-dimensional funnels overlap, since each branch's shape program (27) is solved around its own aim and nothing ties the two solutions together unless this check requires it. There is accordingly nothing to force in advance – either the check passes, in which case the funnels already overlap and no further machinery is needed, or it fails, in which case the segment is re-seeded or rejected exactly as for any other pair of branches, sharing a polytope or not.

Geometric overlap does not imply joint control feasibility; every retained operational r -branch portion is therefore accepted only after the compatibility test of Proposition 3.

F. Graph Construction Algorithm

Algorithm 1 separates geometric route proposals from full-state certification. It retains a branch only when its entry, dynamics, and every handoff pass their checks. A failed terminal attachment does not prevent further geometric extension of a still-valid path. The loop is run per target; accepted branches are then subjected to the cross-branch contingency and compatibility tests of Sec. III-E.

a) *Nominal and certifying control*: The local feedback $-K_i(x - \hat{x}_i)$ witnesses feasibility of each funnel; the command actually applied may instead come from a nominal controller, with the filter enforcing the certified inequalities. Finite-transfer claims require the relevant primary decay condition to remain enforced between certified handoffs.

IV. PROPOSED COMBINATORIAL FILTERING

Assumption 1 (fixed graph/model). Between discrete update events the obstacle map, certified graph, route table, and model used by the filter are fixed; Sec. V treats violations of this assumption.

The graph turns contingency into a nested membership condition: the state must lie in a usable stone of at least r distinct target branches. The geometric filter below preserves that condition. The time-based filter of Sec. IV-B then restricts

Algorithm 1 Directed graph growth with full-state shape optimization

Require: Ω_f ; initial state $x(t_0)$; target j ; terminal certificate $(\hat{x}_{j,T}, P_{j,T}, \rho_{j,T}, k_{j,T})$; margins $\zeta, \gamma_g, \gamma_u$; velocity bound $\bar{v}$; search budget

Ensure: $\mathcal{G}_j = (\mathcal{V}_j, \mathcal{A}_j)$ with certificates $\{(\hat{x}_i, H_i, K_i, k_i)\}$, thresholds $\{\varepsilon_{ik}\}$, and J_j ; or failure

- 1: $P_{j,T} \leftarrow P_{j,T} / \rho_{j,T}, \rho_{j,T} \leftarrow 1$ ▷ normalize to (19)
- 2: $c_0 \leftarrow p(t_0)$; build $\mathcal{P}_0$ by the separating planes of Sec. III-B
- 3: $\mathcal{Q} \leftarrow \{(0)\}$ ▷ queue of polytope-index paths $\pi = (i_0, \dots, i_n), i_0 = 0$
- 4: **while** $\mathcal{Q} \neq \emptyset$ and budget remains **do**
- 5: pop $\pi = (i_0, \dots, i_n)$ from $\mathcal{Q}$
- 6: **if** $\mathcal{P}_{i_n} \cap \Pi_{p, \mathcal{F}_{j,T}} \neq \emptyset$ by (21) **then**
- 7: $\hat{x}_{i_m} \leftarrow$ aim by (16) for each edge; $\hat{x}_{i_0} = [p(t_0); 0]$
- 8: **if** $V_{j,T}(\hat{x}_{i_n}) \geq 1 - \zeta$ **then**
- 9: revise $\hat{x}_{i_n}$ or reject π ; skip to the extension step ▷ terminal interior test, (30)
- 10: **end if**
- 11: select a candidate $k_{i_m} > 0$ (using (18) only for a fixed-gain initialization) and $R_{d_{i_m}}$ from d_{i_m} of (13)
- 12: **for** $m = n$ **down to** 0 **do**
- 13: $x_{\text{in}} \leftarrow \hat{x}_{i_{m-1}}$, or $x(t_0)$ if $m = 0$; pick $\eta_{i_m} < 1 - \zeta$ ▷ (25), (26)
- 14: solve (27) for $(H_{i_m}, Y_{i_m}, \sigma)$
- 15: **if** infeasible **then**
- 16: revise geometry, aim, or k_{i_m} , else backtrack; **break**
- 17: **end if**
- 18: $P_{i_m} \leftarrow H_{i_m}^{-1}, K_{i_m} \leftarrow Y_{i_m} H_{i_m}^{-1}, \rho_{i_m} \leftarrow 1$ by (29)
- 19: bisect $\varepsilon \in (0, 1)$ on (31) to obtain $\varepsilon_{i_m i_{m+1}}$
- 20: **if** no feasible ε **then**
- 21: revise or backtrack; **break**
- 22: **end if**
- 23: **end for**
- 24: accept π iff (22) holds at every node and (30) at every edge, including (26)
- 25: **end if**
- 26: **if** π has no accepted terminal connection **then**
- 27: **for** headings d with $\text{LoS}(z, d) = 1$ in (12), in fan (13) then random order **do**
- 28: c_k by (14); build $\mathcal{P}_k$ as in line 2
- 29: push (π, k) onto $\mathcal{Q}$ if $\dim \mathcal{P}_k = 3$ and $c_k \in \text{int } \mathcal{P}_k$ ▷ seeded overlap by (14)
- 30: **end for**
- 31: **end if**
- 32: **end while**
- 33: keep one certificate per distinct exit aim; delete nodes with no certified continuation
- 34: enforce the cross-branch tests of Sec. III-E
- 35: $\bar{\tau}_e$ by (36), $J_j(i_T)$ by (37), J_j by (38)
- 36: **return** $\mathcal{G}_j$, certificates, thresholds, J_j , or report failure

it to branches whose certified worst-case time-to-go fits the remaining deadline. These costs are time bounds, not fuel costs.

A. Nested Geometric Filter

Let p be the number of candidate targets, with target branch $j \in \{1, \dots, p\}$. For a fixed graph, let p_j^{br} be the number of valid stepping stones on branch j , indexed by $i = 1, \dots, p_j^{\text{br}}$, each with a certified successor chain to target j , i.e. $J_j(i) < \infty$ in (38).

Definition 2 (Nested combinatorial reach-avoid region). The usable branch region and the r -of- p region are

$$\mathcal{B}_j = \bigcup_{i=1}^{p_j^{\text{br}}} \mathcal{F}_{j,i}, \quad \mathcal{C}_r^{\text{geo}} = \{x : \#\{j : x \in \mathcal{B}_j\} \geq r\}. \quad (41)$$

$\mathcal{B}_j$ is a union precisely because remaining on branch j means occupying *any one* of its stones, not a fixed one: the vehicle might be at stone 2 or stone 7 of target C 's chain, either one still certifies a continuation to C . Remaining on branch j is a 1-of- p_j^{br} condition; preserving contingency is an r -of- p condition across the branch regions. This is Definition 3 of [1] with the single certified region $\mathcal{R}_j$ replaced by the union $\mathcal{B}_j$ – the only structural change the graph introduces.

For each nonempty branch, define

$$\begin{aligned} h_{j,i}(x) &= 1 - \frac{V_{j,i}(x)}{\rho_{j,i}}, & b_j(x) &= \max_{1 \leq i \leq p_j^{\text{br}}} h_{j,i}(x), \\ \tilde{b}_r(x) &= \max_{j: p_j^{\text{br}} > 0}^{(r)} b_j(x). \end{aligned} \quad (42)$$

Then $b_j \geq 0$ iff the state lies in some usable stone of branch j , and $\tilde{b}_r \geq 0$ iff at least r distinct target branches are available.

Select a primary cell $(j^\dagger, i^\dagger)$ and write $V^\dagger = V_{j^\dagger, i^\dagger}$ and $h_\dagger = h_{j^\dagger, i^\dagger}$. The filter is the single quadratic program

$$\begin{aligned} \min_{a \in \mathcal{U}, \omega \geq 0} & \quad \frac{1}{2} \|a - a_{\text{nom}}\|_2^2 + p_\omega \omega^2 & (43a) \\ \text{s.t.} & \quad L_f V^\dagger + L_g V^\dagger a \leq -k_{j^\dagger, i^\dagger} V^\dagger + \omega \text{ReLU}(-h_\dagger) & (43b) \\ & \quad L_f h_{j,i} + L_g h_{j,i} a \geq -k_V h_{j,i} - \omega \rho(h_{j,i} - \tilde{b}_r), & (43c) \\ & \quad i = 1, \dots, p_j^{\text{br}}, \quad j = 1, \dots, p. \end{aligned}$$

with $\mathcal{U} = \{a : -\bar{a} \leq a \leq \bar{a}\}$, $p_\omega > 0$, and $\rho : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ positive definite, e.g. $\rho(s) = k_\rho s^2$. The objective (43a) stays minimally invasive on the nominal command; the steering row (43b) drives the primary funnel; the contingency rows (43c) preserve the r -of- p condition. All three share the single slack ω , which is what lets the pivot rows bind while the rest relax.

Remark 11 (Number of primitive certificates and optional pruning). The nested construction does *not* reduce the filter to one constraint per target. Each stone margin $h_{j,i}$ is a primitive certificate, so the geometric QP contains $\sum_{j=1}^p p_j^{\text{br}}$ contingency rows, in addition to the primary steering row. The deadline version similarly contains one row per available deadline-restricted edge certificate, namely $\sum_j |\mathcal{A}_j^D(t)|$. The computational benefit of the nested formulation is therefore not a reduction to p rows, but avoidance of explicit enumeration of the $\binom{p}{r}$ possible target subsets.

If the number of stones becomes large, an optional active-row implementation can be introduced by defining, for each branch, $\mathcal{I}_j(x) := \{i \in \{1, \dots, p_j^{\text{br}}\} : h_{j,i}(x) \geq -\sigma\}$, $\sigma > 0$, and enforcing the contingency row only for $i \in \mathcal{I}_j(x)$. This reduces the QP from $\sum_j p_j^{\text{br}}$ primitive rows to $\sum_j |\mathcal{I}_j(x)|$ rows by omitting stones whose margins are far below the current state. In the ideal continuous-time setting this pruning preserves the invariance argument: every pivot-realizing primitive satisfies $h_{j,i} = b_j = \tilde{b}_r$, and on the safe set $\tilde{b}_r \geq 0$, so every such primitive necessarily belongs to $\mathcal{I}_j(x)$. Moreover, a previously inactive primitive must cross the threshold $-\sigma$ before it can become pivot-realizing. In a sampled implementation, however, this argument is not

sufficient by itself, since an omitted primitive could cross the activation threshold between QP updates.

Theorem 1 (Nested combinatorial invariance). *Consider (1) with a fixed graph, exact model, and funnels satisfying (22). Suppose (a, ω) is continuous on each interval between finitely many primary handoffs, with $a(x) \in \mathcal{U}$ and $\omega(x) \geq 0$, and satisfies (43c) for every primitive stone margin on a neighborhood $\mathcal{D} \supset \mathcal{C}_r^{\text{geo}}$. Then $\mathcal{C}_r^{\text{geo}}$ is forward invariant on every interval on which these hypotheses continue to hold: if $x(t_0) \in \mathcal{C}_r^{\text{geo}}$, then $x(t) \in \mathcal{C}_r^{\text{geo}}$ throughout that interval, so at least r target branches remain available (their identities may change with time).*

Proof. For each target j , $b_j = \max_i h_{j,i}$ is the 1-out-of- p_j^{br} composition of its primitive stone margins, while $\tilde{b}_r = \max_j^{(r)} b_j$ is the r -out-of- p composition of the resulting branch margins. Hence $\tilde{b}_r$ is a nested combinatorial barrier function of the form considered in Theorem 2 of [2]. Taking $\alpha(s) = k_V s$ and $\theta = \omega$, (43c) is precisely the corresponding primitive constraint. Therefore, Theorem 2 of [2] implies forward invariance of $x : \tilde{b}_r(x) \geq 0 = \mathcal{C}_r^{\text{geo}}$ on each interval between primary handoffs. At a primary handoff, only (43b) changes; the state, the primitive contingency margins, and $\tilde{b}_r$ are unchanged. The interval-wise invariance results therefore concatenate across the finitely many handoffs. $\square$

Remark 12 (Contingency is preserved only while requested). Theorem 1 is conditional on joint feasibility of the r -branch rows. For separated targets, the branch regions may eventually diverge, so an r -fold overlap need not persist all the way to a terminal site. In operation the geometric filter is therefore used over the portion of the graph on which the r -of- p tunnel is to be preserved; after a commitment event the requirement is reduced to the selected branch, whose progress is supplied by (43b) and Proposition 2.

Proposition 3 (Sufficient QP regularity condition). *Fix a primary cell and an operational region $\mathcal{D}_{\text{op}} \subset \mathcal{C}_r^{\text{geo}}$ containing no primary handoff, and let $\mathcal{Z}_r(x) = \{(j, i) : h_{j,i}(x) = \tilde{b}_r(x)\}$ be the rows whose slack coefficient $\rho(h_{j,i} - \tilde{b}_r)$ vanishes. Suppose that at every $x \in \mathcal{D}_{\text{op}}$ there exists $a \in \text{int } \mathcal{U}$ satisfying strictly every row in $\mathcal{Z}_r(x)$, and (43b) when $h_\dagger \geq 0$. Then (43) satisfies Slater locally on $\mathcal{D}_{\text{op}}$ and, for the fixed primary cell, its unique optimizer is continuous there.*

Proof. The proof follows Proposition 1 of [1]. For every $(j, i) \notin \mathcal{Z}_r(x)$, positive definiteness of ρ gives $\rho(h_{j,i} - \tilde{b}_r) > 0$. Hence a sufficiently large shared ω makes all such finitely many rows strict. For $(j, i) \in \mathcal{Z}_r(x)$ the relaxation term vanishes, and strict feasibility is provided by the assumed input a . Likewise, the primary row is relaxed by ω when $h_\dagger < 0$, while for $h_\dagger \geq 0$ it is strict by assumption. Thus the pointwise Slater condition used in Proposition 1 of [1] holds, and the same argument yields local continuity of the QP optimizer on $\mathcal{D}_{\text{op}}$. $\square$

B. Deadline-Restricted Time-Based Filter

The geometric filter preserves routes without testing whether they fit the remaining time budget. Let T_D be the mission deadline. Rather than prescribing a commitment time in advance, we preserve at least r deadline-feasible target options until a commitment event at time $\tau_C < T_D$. The time τ_C may be selected online by the mission or planning logic. At commitment, one of the currently certified branches is selected; after that event only the selected branch must be maintained, and its chained decay certificate guarantees completion by T_D .

For a certified edge $e = (i, k)$ on branch j , let $B_{j,e}(t) = T_D - t - J_j(k)$ be the time available to complete the current edge after reserving the certified time $J_j(k)$ required for the remaining suffix. When $B_{j,e} \geq 0$, the current-edge bound of Proposition 1 gives

$$\tau_{ik}(x) \leq B_{j,e} \iff V_{j,i} \leq \varepsilon_{ik} e^{k_{j,i} B_{j,e}}. \quad (44)$$

Define the allowed level and its normalized margin by

$$s_{j,e}(t) = \min\{\rho_{j,i}, \varepsilon_{ik} e^{k_{j,i} B_{j,e}(t)}\}, \quad h_{j,e}^D(x, t) = 1 - \frac{V_{j,i}(x)}{s_{j,e}(t)}. \quad (45)$$

Then $h_{j,e}^D \geq 0$ certifies, in one inequality, both membership in the full-state funnel and sufficient remaining time for this route. Early in the mission the full funnel is available ($s_{j,e} = \rho_{j,i} \equiv 1$ by the normalization of Sec. III-C); as the latest departure time approaches, the allowed level shrinks toward the handoff threshold.

We refer to the pair consisting of an edge $e = (i, k)$ and its time-varying deadline certificate $h_{j,e}^D(x, t)$ as an *edge clock*. The terminology reflects that the admissible portion of the source funnel shrinks as the remaining edge budget $B_{j,e}(t)$ decreases; the clock is not an additional dynamical state.

An edge clock is available only while $B_{j,e}(t) \geq 0$ and $J_j(k) < \infty$. If $B_{j,e}(t) < 0$, even an instantaneous handoff would leave insufficient certified time for the suffix from k to target j . Let $\mathcal{A}_j^D(t)$ collect the available edge clocks, and define the inner and outer pivots exactly as in (42),

$$b_j^D(x, t) = \max_{e \in \mathcal{A}_j^D(t)} h_{j,e}^D(x, t), \quad \tilde{b}_r^D(x, t) = \max_j^{(r)} b_j^D(x, t), \quad (46)$$

with empty branches excluded. The certified region is the time-varying set $\mathcal{C}_r^D(t) = \{x : \tilde{b}_r^D(x, t) \geq 0\}$: at least one time-feasible stone-to-target route remains available for each of at least r distinct targets.

For interpretation and for the discrete checks below, define the target-level spare-time margin

$$H_j(x, t) := T_D - t - T_j^{\text{cert}}(x). \quad (47)$$

Thus $H_j \geq 0$ exactly when at least one currently attached certified route to target j fits within the remaining time $T_D - t$. By (44), this is equivalent to $b_j^D \geq 0$ for the same route family. Hence $\tilde{b}_r^D \geq 0$ is precisely the condition that at least r targets have nonnegative certified spare-time margin.

For stone i of branch j , $k_{j,i} > 0$ denotes its certified CLF decay rate, defined by $\dot{V}_{j,i} \leq -k_{j,i} V_{j,i}$. The allowed level is constant while capped at $\rho_{j,i}$ and otherwise satisfies $\dot{s}_{j,e} = -k_{j,i} s_{j,e}$. Hence, away from the cap-change instant,

$$\kappa_{j,e}(t) := -\frac{\dot{s}_{j,e}}{s_{j,e}} = \begin{cases} 0, & s_{j,e} = \rho_{j,i}, \\ k_{j,i}, & s_{j,e} < \rho_{j,i}. \end{cases} \quad (48)$$

The total derivative is therefore

$$\dot{h}_{j,e}^D = -\frac{L_f V_{j,i} + L_g V_{j,i} a + \kappa_{j,e} V_{j,i}}{s_{j,e}}. \quad (49)$$

Since $s_{j,e} \geq \varepsilon_{ik} > 0$ while the clock is available, this row is well defined throughout its lifetime.

The deadline filter is then the same program with the margins replaced,

$$\min_{a \in \mathcal{U}, \omega \geq 0} \frac{1}{2} \|a - a_{\text{nom}}\|_2^2 + p_\omega \|\omega\|^2 \quad (50a)$$

$$\text{s.t. } L_f V^\dagger + L_g V^\dagger a \leq -k_{j^\dagger, i^\dagger} V^\dagger + \omega_1 \text{ReLU}(-h_{j^\dagger}) \quad (50b)$$

$$\begin{aligned} \dot{h}_{j,e}^D &\geq -k_V h_{j,e}^D - \omega_2 \rho(h_{j,e}^D - \tilde{b}_r^D), \\ e &\in \mathcal{A}_j^D(t), \quad j = 1, \dots, p, \end{aligned} \quad (50c)$$

with $\omega = (\omega_1, \omega_2) \geq 0$ and (49) supplying the left side of (50c). The primary row retains the original geometric margin $h_{j^\dagger}$ from (42), so the steer-enter-overlap-switch logic of Sec. III-C is unchanged. The deadline-restricted margin is used only in the contingency rows.

a) *Relation to [1]'s reach-avoid filter*: The construction follows the time-varying combinatorial reach-avoid filter of [1], but uses the explicit certificate $h_{j,e}^D$ in place of a finite-horizon value function. Here $h_{j,e}^D$ is piecewise C^1 : its derivative changes only at finitely many cap transitions, while the instants at which clocks run out of time are handled by Lemma 3. Hence the invariance argument of [1] applies on each event-free interval and concatenates across these events.

Lemma 2 (Scheduled events). *Between graph updates, each clock has at most one cap change, at $t = T_D - J_j(k) - k_{j,i}^{-1} \log(\rho_{j,i}/\varepsilon_{ik})$, and can run out of time only once, at $t = T_D - J_j(k)$. Both are computable offline from T_D and the route table, and there are finitely many of them in $[0, T_D]$ (hence also prior to commitment).*

Proof. $B_{j,e}$ is affine and strictly decreasing in t , and $s_{j,e}$ is capped exactly while $\varepsilon_{ik} e^{k_{j,i} B_{j,e}} \geq \rho_{j,i}$. $\square$

Lemma 2 only establishes that clock changes occur at finitely many known times. It does not yet show what happens when a clock runs out of time. If, at that instant, no other clock on the same branch certifies the current state, then b_j^D and hence $\tilde{b}_r^D$ could drop instantaneously. The next lemma shows that this cannot occur.

Lemma 3 (Clock handoff). *Let i , ℓ , and m denote three consecutive stones on branch j , with $i \rightarrow \ell \rightarrow m$, and let (ℓ, m) attain the Bellman minimum at ℓ . If $h_{j,(i,\ell)}^D \geq 0$ when*

clock (i, ℓ) runs out of time, then $h_{j,(\ell,m)}^D \geq 0$ at the same instant.

Proof. When (i, ℓ) runs out of time, $B_{j,(i,\ell)} = 0$, so $s_{j,(i,\ell)} = \varepsilon_{i\ell}$. Hence $h_{j,(i,\ell)}^D \geq 0$ implies $V_{j,i} \leq \varepsilon_{i\ell}$ and therefore $x \in \mathcal{F}_{j,\ell}$ by (30). At the same instant,

$$B_{j,(\ell,m)} = J_j(\ell) - J_j(m) = \bar{\tau}_{\ell m},$$

where the second equality follows from (38). By (36), this gives $s_{j,(\ell,m)} = \rho_{j,\ell}$. Thus $x \in \mathcal{F}_{j,\ell}$ implies $h_{j,(\ell,m)}^D \geq 0$. $\square$

The terminal funnel-to-sink clock runs out of time only at T_D and needs no successor because the r -of- p requirement ends at commitment, with $\tau_C < T_D$.

Theorem 2 (Deadline-restricted combinatorial invariance). *Fix the graph, route table, and model. Suppose (a, ω) is continuous on each event-free interval, with $a \in \mathcal{U}$, and satisfies (50c) for every available clock. If $x(t_0) \in \mathcal{C}_r^D(t_0)$, then $x(t) \in \mathcal{C}_r^D(t)$ until commitment at $\tau_C < T_D$. Consequently, at least r targets retain a certified route that can reach the target by T_D . If one such route is selected at τ_C and its chained decay conditions are thereafter enforced, the selected target is reached by T_D .*

Proof. On each event-free interval, the primitive margins $h_{j,e}^D$ are C^1 and (50c) has the combinatorial barrier form of [2]. Applying the nested invariance result of [2] to the time-augmented system, as in the reach-avoid construction of [1], preserves $\tilde{b}_r^D \geq 0$ on that interval. At a cap change, $h_{j,e}^D$ is continuous, while Lemma 3 preserves branch availability when a clock runs out of time. Since Lemma 2 gives only finitely many such events, these interval-wise guarantees concatenate up to τ_C .

At commitment, $\tilde{b}_r^D \geq 0$ certifies at least r deadline-feasible branches. For any selected branch, its certified remaining-time bound fits within $T_D - \tau_C$; therefore Proposition 2 gives arrival at the target by T_D . $\square$

b) Progress: The primary steering row provides continuous progress along the selected branch. The deadline rows instead ensure that retained contingency branches remain feasible with respect to the shrinking time budget. While $s_{j,e} = \rho_{j,i}$ they reduce to geometric membership conditions. Once $s_{j,e}$ begins to shrink, maintaining $h_{j,e}^D \geq 0$ requires $V_{j,i}$ to decrease accordingly; as $B_{j,e} \rightarrow 0$, this condition becomes $V_{j,i} \leq \varepsilon_{ik}$. Hence any branch that is to remain deadline-feasible must complete its handoff by the corresponding deadline.

C. Online implementation

At each sample, query the containing valid funnels, update the available edge clocks and handoffs, evaluate the relevant nested pivot, and solve either the geometric QP (43) or the deadline QP (50). A pointwise QP solve or endpoint-only lookahead does not establish intersample safety; the held command must be validated over the sampling interval. Any graph or model update invokes the restart procedure of Sec. V before full filtering resumes.

V. GRAPH UPDATES

The preceding guarantees assume a fixed obstacle map, graph, route table, and model. If a new obstacle is detected at time t_m , let the free space change from Ω_f^- to Ω_f^+ and remove every stone whose position projection is no longer certified collision-free,

$$\mathcal{V}_{\text{bad}} = \{(j, i) : \Pi_p \mathcal{F}_{j,i} \not\subset \Omega_f^+\}, \quad (51)$$

together with its incident edges.

A. Effect of an Update

The fixed-graph hypotheses of Theorems 1 and 2 no longer hold after t_m . Moreover, any cached $J_j(i)$ whose minimizing route used a deleted edge may underestimate the new time-to-go and must therefore be recomputed. The combinatorial margin may also change discontinuously even though the state does not: if all current attachments to target j are invalidated, then

$$T_j^{\text{cert},+}(x(t_m)) = +\infty, \quad H_j^+(x(t_m), t_m) = -\infty.$$

Such a jump cannot be handled by the continuous barrier inequalities, which constrain derivatives rather than discrete changes.

This differs from the scheduled cap changes and clock handoffs of Sec. IV-B: those events are part of the fixed certificate construction, whereas an obstacle update changes the graph itself.

B. Repair and Restart

For a pure map update, the local CLF certificate of every retained stone remains valid; what may be lost is collision-free validity of deleted stones or their certified continuation to a target. By contrast, a model change, actuation degradation, or certificate update may also invalidate the local decay-feasibility condition $\mathcal{U}_{j,i}(x) \neq \emptyset$, which must then be rechecked before restarting the filter.

Since deleting edges removes options from (38), the updated values satisfy

$$J_j^+(i) \geq J_j^-(i)$$

at affected nodes and can be recovered by a backward update through their predecessors.

For an update before commitment, recompute the affected route-table values, attachments, and deadline clocks. The deadline guarantee can restart only if

$$\sum_{j=1}^p \mathbf{1}[T_j^{\text{cert},+}(x(t_m)) \leq T_D - t_m] \geq r, \quad (52)$$

and the updated filter again satisfies the feasibility and regularity hypotheses of Theorem 2. Under these conditions, Theorem 2 restarts from t_m using the updated graph as the new fixed baseline.

If (52) fails, the requested r -fold contingency is no longer certified and invalidated routes cannot be reused. Continued operation then requires a separately certified fallback. After commitment, the same check is required only for the selected branch.

VI. CONCLUSION

A directed graph of locally certified Lyapunov funnels extends steering-with-contingency from target-centered regions to distant initial states. The growth program jointly certifies full-state funnel geometry, admissible feedback, entry containment, and finite-time handoffs; the cached edge bounds then provide certified target time-to-go values. A nested combinatorial filter preserves membership in at least r target branches, and shrinking deadline-restricted levels strengthen this to r routes whose worst-case certified completion times fit the remaining budget while also forcing progress. The guarantees are conditional on joint feasibility of the critical rows and on graph validity. Map or model updates are therefore handled as discrete events requiring pruning, recomputation, and restart checks rather than being absorbed into the continuous invariance proof.

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